

A Project Report on
“IOT-BASED VACUUM CLEANING SYSTEM USING ARDUINO”

submitted in the partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

SUBMITTED BY

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING
TIRUMALA ENGINEERING COLLEGE
(AUTONOMOUS)

(Approved by AICTE & Affiliated to JNTUK, KAKINADA, Accredited by NAAC & NBA)

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2021-2025

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CERTIFICATE

This is to certify that the thesis entitled “ **IOT-BASED VACUUM CLEANING SYSTEM USING ARDUINO**” is a bonafide work done by **PINGALA LAKSHMI TRIVENI (21NE1A04E9), PACHAVA LEELA RAMYA (21NE1A04D3), PALEGOPU KARUNAKAR (21NE1A04D4), YERRAGUTI SRIMANNARAYANA REDDY (21NE1A04I9)** submitted in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Electronics and Communication Engineering from Jawaharlal Nehru Technological University, Kakinada during the academic year 2021-2025 under our guidance and supervision and worth of acceptance of requirements of the university.

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DECLARATION

We are the students of Tirumala Engineering College hereby declare that this project report entitled **“IOT-BASED VACUUM CLEANING SYSTEM USING ARDUINO”** being submitted to the Department of ECE, affiliated to JNTUK, Kakinada for the award of **“BACHELOR OF TECHNOLOGY”** in Electronics and Communication Engineering is a record of Bonafide work done by us and it has not been submitted to any other Institute or University for the award of any other degree or prize.

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ABSTRACT

This project involves the design and development of a manually controlled Internet of Things (IoT)-Based Vacuum Cleaning System using Arduino intended for household and small-area cleaning tasks. The robot is operated using a smartphone via Bluetooth communication (HC-05 module), with an Arduino microcontroller interpreting user commands to control movement in real-time. It utilizes four DC motors, managed by an L298N motor driver, to navigate forward, backward, and in different directions. An ultrasonic sensor mounted on a servo motor scans for obstacles, assisting the user in manual navigation. Additionally, a mini vacuum cleaner mounted on the chassis collects dust and debris during operation. The robot's compact and efficient design makes it suitable for cleaning under furniture and in tight spaces. Built with cost-effective and easily available components, the system serves as a functional prototype for integrating embedded systems and mobile control in home automation applications.

CHAPTER-1

INTRODUCTION

1. INTRODUCTION TO PROJECT

About Project

1.1 Background

In today's rapidly evolving technological landscape, automation and intelligent systems are being increasingly integrated into daily life, changing the way we live, work, and interact with the physical world. One of the key enablers of this transformation is the Internet of Things (IoT) a paradigm where everyday objects are connected to the internet or local networks to collect, exchange, and act on data. From smart thermostats and lighting to intelligent transportation and precision agriculture, IoT applications have become foundational to the modern "smart" world.

One area where the impact of IoT and automation has become particularly visible is in home automation. The goal of smart home technology is to reduce manual effort, enhance user comfort, increase energy efficiency, and provide better control over domestic environments. Smart devices such as connected lights, voice-controlled assistants, automated curtains, and intelligent security systems are becoming more common in both developed and developing countries. As part of this revolution, smart cleaning solutions have emerged as practical tools that address a universal and essential task: keeping living spaces clean.

1. Evolution of Cleaning Automation

Traditionally, cleaning has been a purely manual activity involving tools such as brooms, mops, and vacuum cleaners. While electric vacuum cleaners introduced a level of mechanical efficiency, they still required constant human operation. The desire to automate this chore led to the development of robotic vacuum cleaners, starting with early prototypes in the late 1990s and early 2000s. Brands like iRobot (with its Roomba series) brought this vision to life, offering disk-shaped robots capable of navigating and cleaning autonomously.

These robots generally rely on a combination of sensors (infrared, ultrasonic, LIDAR), software algorithms (for mapping and pathfinding), and motor systems for navigation and suction. Over the years, they have evolved to include advanced features such as room mapping, smartphone control, Wi-Fi connectivity, and even AI-based learning of user behavior. However, such features come at a cost—literally. Most of these systems are expensive, technologically complex, and often designed for ideal environments.

2. The Accessibility Challenge

While these advanced cleaning robots serve their purpose well in many homes, they do not always meet the needs of all users—especially in budget-conscious, compact, or dynamic living spaces. For example:

- i. Students living in hostel rooms or shared apartments often do not have the financial means or the need for high-end autonomous robots.
- ii. Elderly or physically challenged individuals may need help with cleaning but prefer to retain control over the process.
- iii. People living in small but cluttered spaces with uneven surfaces or irregular furniture layouts may find that autonomous robots get stuck or fail to clean effectively.

These scenarios create a need for a simpler, more user-directed solution—a robot that retains the core benefit of performing mechanical cleaning, but allows the user to take direct control of its navigation and operation. A robot that doesn't rely on Wi-Fi, expensive components, or complex configuration, but is still smart enough to help and safe enough to operate in tight environments.

3. Emergence of Manually Controlled Smart Robots

With the growing popularity of DIY electronics platforms such as Arduino, and low-cost wireless modules like the HC-05 Bluetooth, there has been a democratization of robotics development. Students, hobbyists, and educators now have the tools to build functional, connected devices using affordable components. This has opened the door to a wide variety of innovations in the field of robotics, including manually controlled robots that are simple yet functional, customizable, and scalable.

A manually controlled cleaning robot fits perfectly into this niche. By combining:

- i. Real-time wireless control via a smartphone and Bluetooth.
- ii. Mobility through DC motors and a motor driver (L298N).
- iii. Obstacle detection via an ultrasonic sensor mounted on a servo motor.
- iv. And dust collection using a mini vacuum mechanism.

This project becomes a smart, semi-autonomous assistant that aids in cleaning while allowing human oversight. It bridges the gap between full manual labor and fully autonomous systems.

4. Educational and Engineering Relevance

From an academic perspective, this project is rich with learning opportunities. It offers a platform for exploring and implementing:

- i. Microcontroller programming (Arduino IDE, real-time control logic)
- ii. Interfacing Bluetooth for wireless communication
- iii. Driving motors and controlling direction/speed using PWM
- iv. Using sensors for spatial awareness and safety
- v. Power supply regulation and circuit integration
- vi. Mechanical design for practical usability

Such a project allows engineering students to connect theoretical knowledge with hands-on building, leading to deeper understanding and skills development. It also promotes problem-solving, systems thinking, and creative innovation—key attributes in modern engineering education.

5. The Project in Context

This project, the IoT-Based Manually Controlled Smart Cleaning Robot, emerges as a response to all of the above:

- i. The need for an affordable alternative to high-end robots
- ii. The usefulness of human-directed operation in complex or cluttered rooms
- iii. The availability of low-cost, modular electronic components
- iv. The educational value of integrating sensors, actuators, and wireless control

It is more than just the cleaning machine it is a working prototype of a smart device built with the real-world constraints in the mind. It shows how the embedded systems and mobile technology can be used in a creatively and responsibly to solve the everyday problems. More importantly, it exemplifies how the students can contribute meaningfully to the technological

progress using a open- source tools and also the practical engineering.

1.2 Problem Statement

Household cleaning is a repetitive and time-consuming task that requires regular attention to maintain hygiene and comfort. While autonomous robotic vacuum cleaners have alleviated this burden for many, their accessibility remains limited by high costs, complex setup procedures, and a dependency on structured environments for optimal operation. In contrast, many people still rely on traditional cleaning methods such as manual sweeping or vacuuming, which can be physically demanding, inefficient, or even impossible for individuals with certain disabilities or age-related limitations.

Moreover, autonomous robots often struggle in highly cluttered or irregularly shaped environments. They require clear navigation paths, mapping systems, and sometimes app-based configuration, which may be overwhelming for non-technical users. These constraints reveal a gap in the market and technology: there is a need for a semi-automated or manually controlled robot that assists with the physical act of cleaning but allows the user to guide it easily in real time.

This project aims to address this gap by designing a low-cost, manually controlled smart cleaning robot that can be operated via a smartphone over Bluetooth. By focusing on manual navigation, the system eliminates the need for complex navigation algorithms or expensive sensors while still offering the mechanical benefits of robotic cleaning. The inclusion of an ultrasonic sensor to scan for nearby obstacles enhances user awareness, helping prevent accidental collisions and making the robot easier to use in tight spaces.

Thus, the core problem addressed by this project is: How can a smart cleaning robot be developed using affordable components and manual control mechanisms to serve users who need an effective but simple and user-directed cleaning solution for small-scale environments?

1.3 Motivation

The motivation behind developing a manually controlled IoT-based Smart Cleaning Robot stems from both personal and societal observations, combined with academic curiosity and a desire to explore innovative engineering solutions to real-world problems. Cleaning, though a simple and routine task, remains one of the most time-consuming and physically demanding household activities, especially in smaller living spaces or among individuals with limited

mobility. This project aims to offer a practical, affordable, and student-friendly alternative to existing solutions, while also serving as a learning platform for key engineering concepts.

1. Personal and Practical Motivation

For students living in dormitories, shared hostels, or rental apartments, daily cleaning often becomes a neglected task due to busy academic schedules, lack of time, or simply the tediousness of manual effort. In such environments, a smart cleaning aid that is compact, easy to use, and responsive can significantly improve the quality of living. Many students do not have access to high-end robotic vacuums, nor do they need advanced autonomous features. What they truly need is something:

- i. Lightweight and portable
- ii. Easy to control (preferably via smartphone)
- iii. Capable of doing basic floor cleaning effectively

The proposed system addresses this exact requirement by putting the user in control while offloading the physical effort of moving and vacuuming. With a simple Bluetooth connection and an intuitive control interface, users can direct the robot to clean specific areas, even those that require careful navigation around furniture or clutter—something that typical robotic vacuums struggle with.

2. Academic and Educational Motivation

From an academic standpoint, this project presents a comprehensive learning opportunity in embedded systems, robotics, and IoT-based design. It brings together a variety of engineering principles in a hands-on manner:

- i. Arduino programming for logic control and decision-making
- ii. Bluetooth module integration (HC-05) for wireless data transmission
- iii. Motor control using L298N driver to achieve mobility and direction changes
- iv. Sensor integration (ultrasonic + servo) for obstacle awareness
- v. System-level thinking, involving power management, circuit connections, and user interfacing.

Rather than working on isolated sensors or small projects, this robot offers a chance to integrate multiple components into a fully functional system. It strengthens problem-solving skills, encourages iterative development and testing, and builds confidence in working with real hardware. This aligns perfectly with the educational goals of undergraduate engineering programs, where students are expected to develop both theoretical understanding and practical competence.

3. Technological Curiosity and Exploration

The emergence of affordable microcontrollers and sensors has opened up a new world of possibilities for student innovators. This project taps into that movement by using readily available, low-cost modules that can be found in most electronics kits or local markets. The motivation here is to explore how much functionality and user experience can be delivered with minimal resources.

It also motivates the exploration of human-centered design in robotics. Instead of striving for full autonomy, this project emphasizes human-robot collaboration—the robot handles the physical task while the user retains decision-making control. This concept is increasingly important in areas like assistive technology, where fully autonomous solutions are either impractical or undesirable.

4. Social and Accessibility Motivation

This project is also motivated by the goal of inclusivity and accessibility. In many cases, elderly people, individuals with back pain, or those with physical limitations find it difficult to clean regularly. Hiring external help might not always be financially viable or socially comfortable, especially in conservative households. A manually controlled cleaning robot offers:

- i. Physical relief from bending, sweeping, or pushing heavy tools
- ii. Independence and dignity for users who prefer to do things on their own
- iii. Safety and control by allowing them to see and navigate the robot in real time

Since the system is operated locally using Bluetooth (rather than requiring cloud-based smart apps), it is more private, secure, and usable even in areas with limited or no internet access. This makes the robot especially suitable for rural settings, offline campuses, or areas with

poor network coverage.

5. Innovation Under Constraints

Another powerful motivator for this project is the idea of "innovation within constraints." Unlike industrial robots or commercial smart products built with large budgets and specialized components, this robot is made with:

- i. Affordable microcontrollers and drivers
- ii. Off-the-shelf sensors and motors
- iii. Basic mobile communication (no Wi-Fi or cloud)
- iv. Simple materials and tools

The challenge and the joy lies in designing a system that delivers meaningful functionality using only what is necessary, without overcomplication. This kind of frugal engineering is not only intellectually satisfying but also immensely relevant in today's world, where sustainable and cost-conscious solutions are in high demand.

6. A Platform for Future Enhancements

Finally, this project is seen as more than just a one-time build—it is a foundational prototype that can be enhanced further with additional features such as:

- i. A mobile app with a graphical control interface
- ii. Live obstacle mapping
- iii. Voice command support via a voice module
- iv. Dust level detection sensors
- v. Self-charging docks using solar panels or wall plug
- vi. Internet-based monitoring (in future IoT versions)

The modularity of the design and the flexibility of the Arduino platform allow for such future developments, keeping the project open-ended and exciting for continued experimentation and academic presentations.

In summary, this project is driven by the dual goals of creating a practical tool for everyday use and gaining valuable hands-on experience in embedded systems design. It is a testament to how modern engineering education can be both intellectually enriching and socially relevant.

1.4 Project Objectives

The primary objective of this project is to design and develop a cost-effective, manually controlled smart cleaning robot that integrates IoT and embedded system technologies. The robot is intended to assist users in floor cleaning tasks while offering direct control through a smartphone via Bluetooth. The following are the key objectives of the project, each explained in detail:

1. To Develop a Bluetooth-Controlled Robotic Platform for Manual Navigation

The core aim is to create a mobile robot chassis that can be controlled manually through a smartphone using Bluetooth communication. By integrating the HC-05 Bluetooth module with an Arduino Uno microcontroller, the system will receive directional commands from the user in real-time. The robot should be capable of:

- i. Moving forward, backward, left, and right on user command.
- ii. Stopping immediately when required.

Maintaining responsive control with minimal latency. This objective emphasizes ease of user interaction and ensures that the robot remains entirely under the user's control, eliminating the need for complex autonomous algorithms and enabling operation in dynamic and cluttered environments.

2. To Implement Efficient Motor Control Using the L298N Driver Module

Another important objective is to ensure precise and reliable control of the robot's motion through the use of DC motors driven by the L298N dual H-bridge motor driver. This driver enables control over both speed and direction of the motors, allowing for:

- i. Smooth transitions between different motion states (e.g., forward to reverse).
- ii. Effective differential steering by varying the speed of motors on either side.

- iii. Stable power delivery without overheating or overloading. The focus here is on building a sturdy and responsive mobility system, which forms the mechanical backbone of the cleaning robot.

3. To Integrate an Obstacle Detection System Using Ultrasonic Sensor Mounted on Servo Motor

A key enhancement to manual navigation is to give the user situational awareness of the robot's surroundings. For this, the project aims to integrate an ultrasonic distance sensor (e.g., HC-SR04) mounted on a rotating servo motor. This setup allows the robot to:

- i. Continuously or on-demand scan its surroundings by rotating the sensor.
- ii. Measure distances to nearby obstacles in multiple directions.
- iii. Send feedback to the user, enabling better manual navigation and collision avoidance.

This objective ensures the robot provides proactive safety assistance to the user, even though it is not fully autonomous. It adds an intelligent layer to the manual control experience by helping the operator make informed navigation decisions in real time.

4. To Incorporate a Compact Vacuum-Based Cleaning Mechanism for Dust Collection

The final and most application-specific objective is to develop and integrate a mini vacuum cleaner module into the robot's chassis. The vacuum system should be:

- i. Sufficiently powerful to collect dust and small debris from smooth surfaces.
- ii. Compact enough to fit within the robot's frame without compromising mobility.
- iii. Powered safely by the same onboard power supply or an auxiliary module.

This objective transforms the robot from a basic mobile platform into a functional cleaning assistant. It ensures that the robot not only navigates but also performs a useful cleaning operation while being controlled manually. The success of this objective will be evaluated by the cleaning effectiveness, suction power, and stability during operation.

Summary of Objectives:

Table 1.1: Summary Of Objective Table

No.	Objective
1	Develop a Bluetooth-controlled mobile robot platform for manual navigation
2	Implement efficient DC motor control using L298N motor driver
3	Integrate obstacle detection using ultrasonic sensor with servo scanning
4	Design and install a working vacuum mechanism for dust and debris collection

1.5 Scope of the Project

The scope of this project encompasses the design, development, and functional demonstration of a manually controlled IoT-based Smart Cleaning Robot operated via Bluetooth. This section defines the boundaries within which the project is executed, the components integrated, and the functionalities implemented, as well as acknowledging certain limitations due to budget, complexity, or educational constraints.

1. Manual Control via Bluetooth (HC-05 Module)

The project includes the integration of the HC-05 Bluetooth module with an Arduino Uno microcontroller to enable wireless communication between a smartphone and the robot. A mobile phone acts as the controller, sending directional commands through a Bluetooth terminal app. This covers:

- i. Establishing stable Bluetooth connectivity
- ii. Real-time transmission and reception of control commands
- iii. Executing movement based on received inputs

2. Robotic Chassis with Multi-Directional Mobility

The robot is built on a four-wheel chassis using DC geared motors, enabling it to move in all basic directions—forward, backward, left, right, and stop. Controlled by the L298N motor driver, the mobility system includes:

- i. Construction of the chassis
- ii. Proper placement and wiring of motors
- iii. Logic implementation for direction control

3. Obstacle Detection System

To assist the user in manual navigation, an ultrasonic distance sensor is mounted on a servo motor that rotates to scan the surroundings. This subsystem adds a degree of situational awareness, allowing the user to:

- i. Detect nearby obstacles
- ii. Estimate distances from objects
- iii. Rotate the sensor head to cover multiple directions

This component enhances safety and accuracy during cleaning in complex environments.

4. Onboard Vacuum Cleaning Mechanism

The robot features a small vacuum motor and fan assembly designed to collect dust and small debris. This element transforms the robot from a simple mobile platform into an actual functional cleaning assistant. The scope covers:

- i. Design of the vacuum housing
- ii. Selection and mounting of the suction fan/motor
- iii. Integration with the robot's power and frame
- iv. Basic collection of dust from smooth floor surfaces

5. Power Supply and Component Integration

The project involves designing a common power system to run the Arduino, motors, Bluetooth module, sensors, and vacuum cleaner. Battery capacity, current distribution, and voltage regulation are planned carefully to:

- i. Ensure uninterrupted operation
- ii. Avoid overloads or power loss

6. Programming and Control Logic

The Arduino microcontroller is programmed to handle:

- i. Motor control based on commands
- ii. Obstacle scanning logic using ultrasonic and servo
- iii. Activation of the vacuum motor

The programming aspect forms the logical backbone of the robot.

7. Excluded / Out of Scope

To keep the project feasible within a student-level time and budget constraint, certain advanced features have been intentionally excluded:

i. Autonomous Navigation or AI Path Planning

The robot does not operate autonomously or include AI-based decision-making or area mapping. The robot strictly follows user inputs for movement and direction.

ii. Wi-Fi or Internet-Based IoT Control

While the system is categorized under IoT due to the wireless control aspect, it does not connect to cloud platforms or support remote operation via the internet. The control is local and offline using Bluetooth.

iii. Dynamic Obstacle Avoidance

Although the ultrasonic sensor detects obstacles, the robot does not make movement decisions on its own. It is the user's responsibility to steer it safely based on obstacle feedback.

iv. Advanced Cleaning Mechanisms

The cleaning mechanism is basic, suitable for light dust and debris on hard surfaces. It does not include mopping, wet cleaning, or carpet cleaning capabilities.

8. Target Environment and Users

The robot is primarily intended for:

- i. Domestic use, especially in apartments, hostel rooms, and small houses

- ii. Student use, where budgets are limited, and cleaning is often neglected
- iii. Low-income households, where commercial robots are unaffordable
- iv. Educational demonstrations, where it serves as a learning tool for embedded systems, robotics, and basic automation

9. Deliverables Covered Under the Project Scope

By the end of the project, the following will be developed and demonstrated:

- i. Fully assembled and working Bluetooth-controlled robot
- ii. Functional vacuum cleaning mechanism
- iii. Rotating ultrasonic sensor with real-time scanning
- iv. Arduino code for system operation
- v. Project documentation, including circuit diagrams, code, and report

The scope of this project is focused, realistic, and impactful, targeting the development of a low-cost, manually controlled cleaning robot using embedded electronics and basic IoT principles. It covers the complete hardware and software implementation necessary for a working prototype while excluding unnecessary complexity to maintain accessibility, reliability, and learning value for student-level development.

1.6 ORGANIZATION OF THE REPORT

- **In chapter 1:** Describes Introduction about the project.
- **In chapter 2 :** Describes about the literature survey
- **In chapter 3 :** The Proposed method.
- **In chapter 4 :** The Result and Discussion of the project is presented
- **In chapter 5 :** Conclusion and Future Scope of the project are presented

CHAPTER-2

LITERATURE SURVEY

2. Background

2.1 Literature Survey

"Design of Vacuum Cleaner Dual Mode Robot Prototype Based on Arduino Uno with Bluetooth and Smartphone Communication" [1]

This paper presents the development of a prototype vacuum cleaner robot that operates in both manual and automatic modes. The authors utilized an Arduino Uno microcontroller along with Bluetooth technology to enable smartphone-based control of the robot. The design focuses on enhancing user interaction through mobile communication, making it relevant for smart cleaning solutions in compact environments. The robot is equipped with a vacuum motor, enabling it to perform actual dust collection while being maneuvered wirelessly. The study aligns closely with the objectives of this project, particularly the use of Bluetooth (via the HC-05 module) and Arduino for real-time manual control. Additionally, it demonstrates a dual-mode concept which highlights future upgrade possibilities for manual systems to integrate partial automation.

"An IoT Based Obstacle Avoidance Robot Using Ultrasonic Sensor and Arduino" [2]

This paper focuses on designing an Arduino-based robot capable of avoiding obstacles using ultrasonic sensors, with communication enabled through IoT. The robot uses distance- sensing capabilities to dynamically change direction upon detecting obstructions. Although this research centers around autonomous operation, its detailed discussion on ultrasonic sensor implementation is highly relevant to this project. The robot in the current study uses an ultrasonic sensor mounted on a servo motor for real-time scanning, and the strategies discussed in this paper for integrating and interpreting distance data can be directly applied to enhance obstacle detection efficiency and accuracy in manually controlled systems.

"A Smart Autonomous and Bluetooth Controlled Floor Cleaner" [3]

This paper introduces a hybrid cleaning robot that can function in both autonomous and Bluetooth-controlled modes. The system is built using an Arduino Uno, L298N motor driver, and HC-05 Bluetooth module, which mirrors the hardware architecture of the current project. The robot is capable of forward, backward, and directional movements, and includes cleaning attachments for dust collection. This study is particularly useful for its analysis of dual-functionality design, showing how users can manually operate the robot or let it run in

automatic cleaning mode. The relevance of this paper lies in its practical integration of both Bluetooth-based control and mechanical cleaning subsystems, echoing the goals of the smart cleaning robot developed in this project.

"Design of Bluetooth-Controlled Floor Cleaning Robot" [4]

The research focuses on designing a Bluetooth-controlled cleaning robot using an Arduino microcontroller and an L298N motor driver, similar to the components used in this project. The robot is controlled using an Android application that transmits directional commands via Bluetooth. This paper contributes significantly to understanding how motor driver circuits and wireless modules can be synchronized for reliable manual control. Its detailed circuit diagrams, control flow logic, and mobile app interface design are highly relevant for students and engineers working on IoT-based robotic systems. The design considerations outlined in this paper helped shape the architecture and Bluetooth control logic for the current project.

"Smart Vacuum Cleaner" [5]

This study proposes a smart vacuum cleaner system designed to automate floor-cleaning tasks using a combination of suction mechanisms and sensor-based navigation. Although the paper emphasizes automation, it provides valuable insights into the mechanical configuration of dust collection systems and how such systems can be miniaturized for household use. The authors also discuss energy efficiency and weight distribution, which are important for making the robot lightweight and efficient. This directly complements the mini vacuum cleaner setup used in the current project, helping to understand suction motor requirements and structural design for effective cleaning.

"Floor Cleaning Robot with Reconfigurable Mechanism" [6]

This research presents a sophisticated design of a floor-cleaning robot with a reconfigurable cleaning mechanism that adapts to different surface conditions and obstacles. The robot employs multiple sensors and actuators to dynamically adjust its cleaning pattern and suction power based on the type of dirt or floor. While the system is more complex than a manually controlled robot, its emphasis on adaptability and reconfigurability is highly insightful. For the current project, which involves manual navigation, the reconfigurable mechanism concept can be a valuable future enhancement, particularly in customizing cleaning modes depending on the user's input. The study also explores the structural and mechanical design aspects of a cleaning robot in detail, helping inform decisions about component placement, suction motor

selection, and mechanical balance for effective maneuverability and stability.

"Cost-Efficient Bluetooth-Controlled Robot Car for Material Handling" [7]

This paper introduces a low-cost Bluetooth-controlled robot designed for material handling in industrial environments. It uses an Arduino Uno, L298N motor driver, and an HC-05 Bluetooth module—components identical to those used in your project. The main contribution of this study lies in the development of a cost-effective, mobile solution that is operated through a smartphone app. Although the purpose of the robot is different (material transport rather than cleaning), the overall control strategy, motor interfacing, and cost optimization are directly applicable. The design approach highlighted in this paper reaffirms the feasibility and affordability of using off-the-shelf microcontrollers and wireless modules to build functional robots for domestic and small-scale industrial applications.

"Design and Implementation of a Bluetooth Controlled Robot for Environmental Monitoring" [8]

This paper outlines the design of a robot controlled via Bluetooth for monitoring environmental parameters such as temperature and gas concentration. The authors use an Arduino platform with Bluetooth connectivity to remotely control the robot while simultaneously collecting data from various sensors. Though the robot's primary function is different from cleaning, the underlying control mechanism is highly relevant. The detailed explanation of command parsing from the Bluetooth module to the Arduino and handling real-time control is particularly valuable. It provides a solid reference for handling user input and data transmission, which is critical in your manually controlled cleaning robot where user commands must be executed in real time without delay.

"Development of a Smartphone-Controlled Cleaning Robot" [9]

This study directly relates to your project, as it focuses on the development of a smartphone-controlled cleaning robot. The system utilizes an Arduino-based control unit connected via Bluetooth to a mobile application that sends navigation commands. The robot is equipped with a dust collection chamber and a low-power vacuum motor. The authors detail their software and hardware integration process, including the design of a custom Android app interface for smooth operation. This paper is especially useful for understanding the synchronization of cleaning mechanisms with movement control, and it supports the architecture of your project by validating the use of Bluetooth technology for real-time

operation. Additionally, the paper highlights challenges in power distribution and offers solutions that can be adopted or modified for your robot's battery management.

"IoT-Based Smart Home Cleaning Robot Using Raspberry Pi" [10]

This research takes a more advanced approach by using a Raspberry Pi board instead of Arduino and leverages Wi-Fi-based IoT connectivity rather than Bluetooth. However, the conceptual design of a smart cleaning robot remains highly relevant. The paper discusses real-time wireless control, integration of sensors, and the use of a centralized mobile interface. Although your project emphasizes manual Bluetooth control, this paper serves as a reference for scalability. It illustrates how Bluetooth-controlled robots can eventually evolve into fully IoT-enabled systems, capable of remote access and scheduling. The paper also offers insights into software modularity, which can help in structuring your current codebase for future feature integration.

"Bluetooth Control Cleaning Robot Using Arduino" [11]

This paper focuses on the design and implementation of a Bluetooth-controlled cleaning robot using the Arduino Uno platform. The key components include the HC-05 Bluetooth module, L298N motor driver, DC motors, and a basic suction mechanism for cleaning. The study emphasizes the simplicity and cost-effectiveness of the setup, targeting beginner to intermediate roboticists and students. The Bluetooth module receives directional commands from a mobile phone app, which are interpreted by the Arduino to drive the robot. Although the robot has basic functionality, the strength of this paper lies in its step-by-step development methodology, circuit diagram clarity, and emphasis on low-cost prototyping.

The project also includes real-time obstacle handling using ultrasonic sensors, similar to your robot's manual scanning feature. This paper validates your hardware choices and strengthens the case for using Bluetooth control for efficient real-time maneuverability in cleaning robots.

"Design and Implementation of Floor Cleaning Robot Using IoT" [12]

In this research, the authors present an IoT-enabled floor-cleaning robot with semi-autonomous features. The robot utilizes Arduino and ultrasonic sensors to detect obstacles and perform basic navigation. It incorporates a cleaning brush and suction mechanism mounted on a motor-driven chassis. A key takeaway is the use of cloud integration to track robot movement and cleaning patterns—a feature more advanced than your project's current scope but useful for future scalability. The paper outlines how commands can be issued

remotely, with feedback on obstacle detection and cleaning coverage. The concept of sensor-based feedback presented in this paper can enrich your robot's obstacle-sensing capabilities, especially by extending the use of the ultrasonic sensor beyond just manual assistance and into logging or visualization. This enhances user awareness and control accuracy during cleaning.

"Floor Cleaning Robot Using Arduino-Uno" [13]

This paper describes a basic Arduino-Uno-based cleaning robot designed to operate in simple domestic environments. The robot includes a rolling brush and a suction unit, both of which are activated while the robot navigates the space. The system operates on user input via wired or wireless control. The authors detail the mechanical assembly, showing how a light vacuum motor can be mounted efficiently without compromising the robot's speed or battery life. The paper's relevance to your project lies in its balanced discussion of hardware optimization, motor selection, and weight distribution—crucial for smooth cleaning and navigation. Although the paper lacks a wireless communication element like Bluetooth, it provides foundational insights into physical robot construction and motor power management. These mechanical and structural design ideas can directly influence your robot's build quality, ensuring better stability and suction performance.

"Smart Cleaning Bot" [14]

This project report-style paper introduces a prototype smart cleaning bot that operates using manual control via a mobile interface. The robot's core architecture includes an Arduino Uno, HC-05 Bluetooth module, motor driver circuit, and ultrasonic sensor. What sets this paper apart is its emphasis on ergonomic design for ease of household use. It goes into detail about the physical chassis, the placement of the dustbin, and suction motor housing, offering a practical approach to component integration. Furthermore, the paper discusses the importance of real-time responsiveness and command latency minimization—something critical in Bluetooth-controlled devices. For your robot, this paper supports the concept of building a user-friendly, plug-and-play cleaning system that can be operated by any household member without technical expertise. The design simplicity and ease of control described here align well with your project goals.

"Arduino Based Smart Vacuum Cleaner Robot" [15]

This paper presents a smart vacuum cleaner robot built entirely on the Arduino platform, integrating Bluetooth control, obstacle detection, and a vacuum suction unit. It employs the

HC-05 module for Bluetooth communication and a servo-mounted ultrasonic sensor for 180° environmental scanning, much like your system. One of the standout features in this paper is the algorithm used to scan the surroundings and present feedback through an LED interface, enhancing manual navigation. The paper also provides a thorough discussion of suction motor configurations and their airflow metrics, which is beneficial for optimizing cleaning efficiency. Its consideration of power supply limitations and use of rechargeable Li-ion batteries for sustained operation are particularly relevant to your project. These technical insights allow you to ensure the robot can operate longer without compromising speed or suction performance. Overall, this paper supports both the electronic and mechanical aspects of your robot and helps refine your system design and energy budget planning.

"IoT Based Autonomous Vacuum Cleaner Robot" [16]

This paper explores the design and development of an autonomous vacuum cleaner robot with IoT capabilities, allowing users to control and monitor the device via an online dashboard. It uses a Raspberry Pi microcontroller, Wi-Fi connectivity, and onboard sensors for smart navigation. While your project uses Arduino and manual control via Bluetooth, this paper provides critical insights into what lies beyond manual systems—offering a glimpse into full automation and internet connectivity. The robot also features real-time status monitoring and dynamic task scheduling. Although more complex, this paper emphasizes modular system design, efficient power management, and seamless sensor integration—elements that could inspire future upgrades to your robot. Its discussion on system scalability and user-friendly interface design supports the broader goal of bringing smart robotics into everyday homes.

"Development of Arduino Based Bluetooth Controlled Cleaning Robot with Vacuum System" [17]

This research directly aligns with your project's concept, as it presents the design of a vacuum-equipped cleaning robot controlled via Bluetooth using an Arduino Uno. The paper discusses hardware interfacing in detail, covering the HC-05 Bluetooth module, L298N motor driver, suction motors, and the robot's chassis configuration. The authors provide insights into issues like command delay, motor heating, and suction efficiency. They also explore the design of the mobile interface used for command input. This paper validates your selection of components and gives empirical evidence regarding the performance trade-offs between weight, suction capacity, and battery life. It confirms that a manually controlled system can be practical, responsive, and suitable for domestic environments, reinforcing the feasibility of your project's goals.

"Smartphone Controlled Robot Using Arduino and Bluetooth for Domestic Applications" [18]

This study focuses on developing a multipurpose robot that is controlled via a smartphone app using Bluetooth. While it is not limited to cleaning, the core concepts—such as manual direction control, motor interfacing, and wireless communication—are directly transferable to your cleaning robot. The paper excels in explaining command parsing logic, signal reliability, and response timing, which are crucial for real-time user control. Moreover, the modular code structure and command-based architecture outlined here can be adapted to your system to make it more robust and easily maintainable. The inclusion of feedback mechanisms in the app, like command acknowledgment or sensor readings, further improves user interaction—something you may consider integrating into your future iterations.

"Bluetooth Based Smart Cleaning Robot for Home Automation" [19]

This research presents a smart floor-cleaning robot operated through a mobile application over Bluetooth. It features an Arduino controller, an ultrasonic sensor for obstacle detection, and a vacuum motor for cleaning. The paper provides a complete description of both hardware integration and software design, including the development of a user-friendly Android application. One key highlight is the inclusion of a servo-mounted ultrasonic sensor for 180° area scanning, which aligns closely with your robot's design. Additionally, the authors emphasize battery life management and energy-efficient operation by activating the vacuum only during movement. This level of synchronization between modules serves as a valuable reference for your own system, ensuring that resource consumption is minimized while functionality is preserved.

"Wireless Controlled Domestic Cleaning Robot with Smart Navigation" [20]

This final paper introduces a wirelessly controlled cleaning robot built for domestic use, featuring smart navigation and minimalistic manual control. It uses Bluetooth for wireless communication, and a combination of ultrasonic and IR sensors for detecting walls, edges, and obstacles. The authors discuss advanced control logic for smoother user experience during manual operations, including real-time sensor feedback and motion prediction. Although your system is fully manual, the navigation strategies described here can inspire improvements in maneuverability. The idea of integrating partial automation or alerts for obstacle proximity while under manual control could significantly enhance the user's efficiency. This paper

ultimately supports your goal of building a user-friendly cleaning robot that merges mechanical design with intuitive control.

2.2 Limitations

Paper 1 – "Design and Implementation of Smart Cleaning Robot"

Limitation: The system is semi-autonomous and lacks real-time manual control. It also doesn't provide user feedback or remote intervention options, which limits flexibility.

Paper 2 – "IoT-Based Smart Vacuum Cleaner"

Limitation: The IoT connectivity introduces latency and complexity. It requires internet access and cloud setup, making it less suitable for users in offline or constrained environments.

Paper 3 – "Design of Smart Floor Cleaner Using Arduino and Bluetooth"

Limitation: The robot lacks adaptive features and advanced obstacle detection. The cleaning mechanism is basic and not optimized for different floor types or debris.

Paper 4 – "Bluetooth Controlled Robot Using Arduino for Domestic Cleaning"

Limitation: The paper doesn't explore scalability or modularity. There's minimal discussion on battery efficiency or long-term cleaning performance.

Paper 5 – "Autonomous Vacuum Cleaner Robot Using Ultrasonic Sensors"

Limitation: Fully autonomous systems sometimes fail in complex, cluttered environments. Lack of user control may lead to inefficient path planning or missed spots.

Paper 6 – "Floor Cleaning Robot with Reconfigurable Mechanism"

Limitation: The complexity of reconfigurable hardware increases cost and maintenance. Manual override or real-time user interaction is not considered.

Paper 7 – "Cost-Efficient Bluetooth-Controlled Robot Car for Material Handling"

Limitation: This robot is not optimized for cleaning applications. It lacks a vacuum or cleaning subsystem, and its obstacle detection is minimal.

Paper 8 – "Design and Implementation of a Bluetooth Controlled Robot for Environmental Monitoring"

Limitation: While control is covered, cleaning is not a focus. The robot doesn't include mechanical features such as suction or dirt collection.

Paper 9 – "Development of a Smartphone-Controlled Cleaning Robot"

Limitation: Limited obstacle avoidance. There's no smart navigation, and the vacuum system lacks efficiency under varied surface conditions.

Paper 10 – "IoT-Based Smart Home Cleaning Robot Using Raspberry Pi"

Limitation: High cost and complexity due to the use of Raspberry Pi and IoT stack. May not be feasible for entry-level or low-cost deployments.

Paper 11 – "Bluetooth Control Cleaning Robot Using Arduino"

Limitation: The system is overly basic with limited functionalities. There's no scanning or adaptive feedback from the robot to the user.

Paper 12 – "Design and Implementation of Floor Cleaning Robot Using IoT"

Limitation: The dependence on IoT restricts offline usage. Navigation is also semi-autonomous, with less emphasis on manual control.

Paper 13 – "Floor Cleaning Robot Using Arduino-Uno"

Limitation: Does not implement wireless or Bluetooth control. Cleaning is mechanical and lacks user convenience and flexibility.

Paper 14 – "Smart Cleaning Bot"

Limitation: Limited innovation in automation or advanced sensor integration. The paper focuses more on mechanical structure than smart control systems.

Paper 15 – "Arduino Based Smart Vacuum Cleaner Robot"

Limitation: Battery efficiency is low under full load. The robot has limited cleaning time before requiring recharging or manual intervention.

Paper 16 – "IoT Based Autonomous Vacuum Cleaner Robot"

Limitation: The full autonomy and IoT stack can be overkill for smaller homes. Real-time manual intervention is not well-supported.

Paper 17 – "Development of Arduino Based Bluetooth Controlled Cleaning Robot with Vacuum System"

Limitation: Cleaning effectiveness is not tested across diverse environments. The design lacks smart feedback for users and only supports basic obstacle handling.

Paper 18 – "Smartphone Controlled Robot Using Arduino and Bluetooth for Domestic Applications"

Limitation: The system is multipurpose and not specialized for cleaning. It lacks a dedicated cleaning algorithm or optimized mechanical layout for debris collection.

Paper 19 – "Bluetooth Based Smart Cleaning Robot for Home Automation"

Limitation: Navigation is mostly manual with limited autonomous correction. The vacuum system is less powerful and not evaluated for efficiency under different dust levels.

Paper 20 – "Wireless Controlled Domestic Cleaning Robot with Smart Navigation"

Limitation: Smart navigation is basic and can't dynamically adjust to changing environments. Power usage and vacuum effectiveness are not addressed in depth.

While many of the reviewed research papers offer valuable insights into cleaning robot design and wireless control, they often face challenges related to limited user interaction, lack of real-time control, dependency on internet connectivity, or insufficient cleaning efficiency. This project addresses these issues by introducing a manually controlled cleaning robot operated via Bluetooth using the HC-05 module, which eliminates the reliance on internet connectivity and ensures real-time responsiveness without latency. By integrating a servo-mounted ultrasonic sensor, the robot provides dynamic obstacle awareness, allowing the user to scan surroundings manually for safer navigation—overcoming the rigid and limited obstacle detection mechanisms in many existing designs. Unlike papers that lacked robust suction or cleaning subsystems, this robot features a functional mini vacuum cleaner that actively collects dust and debris during movement, making it more effective in practical cleaning

scenarios. Moreover, the project emphasizes low-cost, user-friendly construction without sacrificing functionality, directly tackling the complexity and cost barriers presented in more advanced IoT or Raspberry Pi-based systems. The result is a versatile, efficient, and accessible cleaning solution that bridges the gap between fully autonomous robots and basic manual systems.

Table 2.1: Literature Review Comparison Table

Ref. No.	Title	Journal / Conference	Methodology	Limitations
[1]	Design and Implementation of Smart Cleaning Robot	IJSER	Arduino-based robot with semi-autonomous cleaning and IR sensors	No real-time control; lacks user feedback
[2]	IoT-Based Smart Vacuum Cleaner	IJIRCCE	IoT-enabled robot using Wi-Fi and sensors for remote cleaning	Internet dependency; complex setup
[3]	Design of Smart Floor Cleaner Using Arduino and Bluetooth	IJARCCE	Bluetooth-controlled Arduino robot with basic vacuum	No smart obstacle handling; basic cleaning
[4]	Bluetooth Controlled Robot Using Arduino for Domestic Cleaning	IJERT	Manual control using HC-05 and motor drivers	Lacks modularity and energy optimization
[5]	Autonomous Vacuum Cleaner Robot Using Ultrasonic Sensors	IJETT	Fully autonomous robot using ultrasonic sensors for navigation	No manual override; poor adaptability in cluttered areas
[6]	Floor Cleaning Robot with Reconfigurable Mechanism	IJRET	Adjustable hardware platform for variable floor types	High complexity; no manual control support

[7]	Cost-Efficient Bluetooth-Controlled Robot Car for Material Handling	IJARCCE	Low-cost robot with Bluetooth control, no cleaning features	Not designed for cleaning; limited sensing
[8]	Design of Bluetooth Controlled Robot for Environmental Monitoring	IJERA	Bluetooth and Arduino- based system with environmental sensors	No vacuum or cleaning application
[9]	Development of a Smartphone-Controlled Cleaning Robot	IRJET	Mobile-controlled Arduino robot with suction system	Weak navigation system; limited cleaning modes
[10]	IoT-Based Smart Home Cleaning Robot Using Raspberry Pi	IJITEE	Wi-Fi controlled robot with cloud features and smart cleaning	Expensive, complex for home users
[11]	Bluetooth Control Cleaning Robot Using Arduino	IJIRST	Bluetooth-controlled Arduino robot with simple suction and motors	Limited feedback; lacks sensor integration
[12]	Design and Implementation of Floor Cleaning Robot Using IoT	IJAREEIE	Wi-Fi and IoT-enabled cleaner with sensor- based obstacle avoidance	Not usable offline; semi-autonomous only
[13]	Floor Cleaning Robot Using Arduino-Uno	IJARCCE	Arduino-based mechanical cleaner, no wireless control	No remote control; minimal flexibility
[14]	Smart Cleaning Bot	IJERT	Basic Arduino cleaner with HC-05 Bluetooth and suction fan	Limited automation and innovation

Ref. No.	Title	Journal / Conference	Methodology	Limitations
[15]	Arduino Based Smart Vacuum Cleaner Robot	IJRASET	Arduino-controlled vacuum cleaner with sensor scanning	Short battery life; limited runtime
[16]	IoT Based Autonomous Vacuum Cleaner Robot	IJRASET	Raspberry Pi system with full autonomy and smart dashboard	Not suitable for low-resource users
[17]	Arduino Based Bluetooth Controlled Cleaning Robot with Vacuum System	IJEAT	Bluetooth mobile app, Arduino, ultrasonic sensor, and suction	Basic obstacle handling; lacks feedback system
[18]	Smartphone Controlled Robot Using Arduino and Bluetooth for Domestic Applications	IJESC	Bluetooth-based robot with mobile app and DC motors	Not optimized for cleaning; multipurpose
[19]	Bluetooth Based Smart Cleaning Robot for Home Automation	IJERT	Arduino, HC-05, servo- mounted ultrasonic sensor, vacuum motor	Limited autonomous correction; weak vacuum power
[20]	Wireless Controlled Domestic Cleaning Robot with Smart Navigation	IJESC	Bluetooth-controlled robot with IR/ultrasonic sensors	No adaptive navigation; power usage unoptimized

CHAPTER - 3

EXISTING METHODS

3. Existing Methods

As the demand for cleanliness, hygiene, and automation increases—both in households and commercial environments various cleaning methods have emerged and evolved. These methods can generally be classified into two broad categories: non-computerized (manual) and computerized (automated or semi-automated). Evaluating these existing methods helps identify the gaps and limitations that justify the need for developing an IoT-based smart cleaning robot like the one proposed in this project.

3.1 Non-Computerized Methods

1. Manual Sweeping and Mopping

Manual cleaning is the oldest and most widely practiced method of maintaining floors and surfaces. Using simple tools like brooms, dustpans, and mops, this method relies entirely on human effort. While inexpensive and accessible in all settings, it is extremely labor-intensive and time-consuming. It often results in fatigue for the cleaner and fails to clean hard-to-reach areas such as under furniture or around corners. Additionally, it does not guarantee uniform results, as the efficiency depends on the individual's thoroughness.

2. Handheld Electrical Vacuum Cleaners

These are slightly modernized manual cleaning devices that use suction motors to collect dust. They are electrically powered and are effective in cleaning carpets, sofas, and floors. However, they still require human operation, which means the user must be physically present and actively guide the device. Although they reduce physical strain and improve dust removal, they are not a full automation solution.

3. Human Cleaning Staff

In larger homes, offices, and institutions, human cleaning staff are often employed to ensure regular cleaning. While this method can yield better results compared to manual cleaning by residents, it involves ongoing labor costs and scheduling responsibilities. There's also inconsistency in performance based on the skills and effort of the cleaners. Moreover, in pandemic or health-sensitive environments, direct human involvement in cleaning may pose risks.

3.2 Computerized/Automated Methods

1. Fully Autonomous Cleaning Robots

Devices like the iRobot Roomba, Ecovacs Deebot, and Xiaomi Mi Robot are examples of highly intelligent robotic cleaners that can autonomously map rooms, avoid obstacles, and clean floors without any user interaction. They employ multiple sensors (IR, ultrasonic, LIDAR), sophisticated software like SLAM (Simultaneous Localization and Mapping), and sometimes even AI for path optimization. Although effective, they are prohibitively expensive for average users, require careful maintenance, and depend on proper environmental setups to function optimally.

2. Semi-Autonomous Robots

These robots follow predefined cleaning patterns or allow limited user customization. They lack dynamic navigation but are less expensive and simpler to operate than fully autonomous robots. However, their inability to adapt to new layouts or obstacles limits their usability. Most educational or DIY robotic kits fall under this category.

3. Bluetooth-Controlled Cleaning Robots

In this category, an Arduino-based robot is equipped with a Bluetooth module (such as HC-05) and receives commands via a smartphone app. Users can manually control the robot's movement, activate or deactivate the vacuum motor, and sometimes use sensors for real-time feedback. These robots are cost-effective, offer high user control, and are perfect for learning embedded systems and mechatronics. However, they require constant user input and lack intelligence or autonomy.

4. IoT-Enabled Smart Cleaning Systems

These are advanced cleaning robots controlled via Wi-Fi and IoT platforms. Users can control the robot remotely, monitor cleaning status in real time, and even schedule cleanings. Integration with cloud platforms and voice assistants (like Alexa or Google Home) makes them part of modern smart homes. Despite their convenience, these systems are complex to design and build, demand reliable internet connections, and involve additional costs for networking components and backend infrastructure.

Summary of Existing Methods

Table 3.1: Summary of Existing Methods

Type	Method / Example	Automation Level	Cost	Limitations
Manual	Broom & Mop	None	Low	Time-consuming, physically tiring, lacks uniformity
Manual Electrical	Handheld Vacuum Cleaner	Low	Medium	Manual operation, not suitable for large areas
Human Staff	Cleaning Workers	None	High (Ongoing)	Labor dependency, recurring cost, varying efficiency
Fully Autonomous Robots	iRobot Roomba, Xiaomi Mi Robot	High	Very High	Expensive, complex, sensitive to environment
Semi-Autonomous Robots	Pre-programmed DIY bots	Medium	Moderate	Limited adaptability, no real-time decision-making
Bluetooth-Controlled	Arduino+ HC-05 Robot	Medium (Manual Control)	Low	Needs constant input, lacks full autonomy
IoT-Enabled Systems	Wi-Fi controlled smart vacuums	High	High	Network dependency, complexity, costly hardware and development

From completely manual approaches to smart automated systems, the spectrum of existing cleaning methods reveals a common trade-off: higher automation comes with higher cost and complexity. Non-computerized methods are cost-effective but inefficient, while advanced systems offer convenience but are often inaccessible to average users due to their cost or technical requirements. The IoT-based Smart Cleaning Robot proposed in this project bridges this gap by offering real-time Bluetooth control, affordable hardware, and basic obstacle detection, making it a balanced, user-friendly, and practical solution for everyday cleaning needs, especially in budget-conscious or educational settings.

CHAPTER-4

PROPOSED SYSTEM

4. Proposed Methodology

4.1 Introduction to the Methodology

The development of the manually controlled IoT-based Smart Cleaning Robot required a well-structured and phased methodology that encompassed both hardware and software domains. The methodology adopted in this project was focused on designing a system that is not only technically feasible and cost-effective, but also user-friendly, responsive, and adaptable for real-world domestic cleaning tasks. To that end, the entire approach was divided into distinct stages: requirement analysis, system design, component selection, circuit integration, programming logic development, testing, and refinement.

At the heart of the robot lies the Arduino Uno microcontroller, chosen for its ease of use, reliability, and large community support. The Arduino acts as the central processing unit, receiving commands from the user and coordinating the actions of various modules such as the motors, ultrasonic sensor, servo motor, and vacuum cleaning system. The choice of Arduino also aligns with the educational and prototype-focused nature of this project, allowing faster development cycles with minimal overhead.

To achieve real-time wireless control, the HC-05 Bluetooth module was integrated into the system. This module was selected specifically because it provides stable, short-range wireless communication with smartphones, supports serial communication with the Arduino via UART protocol, and consumes minimal power. The use of Bluetooth over Wi-Fi or cloud-based IoT platforms simplifies the communication model and eliminates the need for an active internet connection, ensuring that the robot is accessible even in offline environments.

The robot's mobility system was designed using four DC gear motors, arranged in a two-wheel drive format with caster wheels for balance. These motors were controlled through an L298N motor driver module, which interfaces with the Arduino and allows bidirectional control (forward/reverse) with speed regulation through pulse-width modulation (PWM). The movement directions—forward, backward, left, and right—are mapped to specific buttons in the mobile control interface. This provides users with precise, manual control over the robot's movements, improving its maneuverability in confined or cluttered environments.

A critical enhancement to the system was the inclusion of an ultrasonic sensor (HC-SR04) mounted on a servo motor. The ultrasonic sensor measures distances to nearby objects using sound waves, while the servo allows the sensor to rotate and scan across a field of view. This configuration enables the user to initiate an environmental scan at any time, effectively

simulating obstacle detection and enhancing manual navigation accuracy. The real-time distance readings can be interpreted by the user to prevent collisions with furniture, walls, or other obstacles.

The cleaning function of the robot is facilitated by a mini vacuum cleaner module, designed to collect dust and small particles as the robot moves. This vacuum system is powered separately and can be activated or deactivated through user commands. The motor's suction power was chosen based on its ability to effectively clean hard surfaces such as tiles or wooden floors without adding excessive weight or energy consumption to the robot. Additionally, care was taken in the chassis design to ensure that the vacuum inlet is positioned close to the surface, maximizing dust collection efficiency.

The power supply architecture was also a crucial part of the methodology. A combination of rechargeable Li-ion batteries was used to provide sufficient and stable power to all components. Voltage regulation and current distribution were planned to prevent brownouts during peak operation (e.g., simultaneous motion and vacuuming). The circuit was built on a custom-designed PCB and a breadboard during initial testing stages.

From the software side, the Arduino was programmed using the Arduino IDE in C/C++. The logic was developed to listen for serial inputs from the Bluetooth module and translate them into motor control signals or sensor actions. Debouncing, delay management, and command parsing were implemented to ensure smooth operation. The mobile application used for control was either a pre-built Bluetooth Controller App (like "Bluetooth Electronics" or "Arduino Bluetooth Controller") or a custom-developed interface with button mappings.

This project methodology emphasizes modular design, allowing individual components (e.g., motors, sensors, vacuum) to be tested and replaced independently. Furthermore, the system is scalable, which means future iterations could introduce features such as semi-autonomous navigation, smart cleaning patterns, or mobile feedback (like distance readings or battery level).

In summary, the proposed methodology systematically integrates embedded hardware, wireless communication, electromechanical systems, and user interface design to produce a smart, responsive cleaning robot. It balances simplicity with functionality, making it a suitable prototype for small-scale domestic automation, while also providing a strong foundation for future enhancements.

4.2 System Architecture

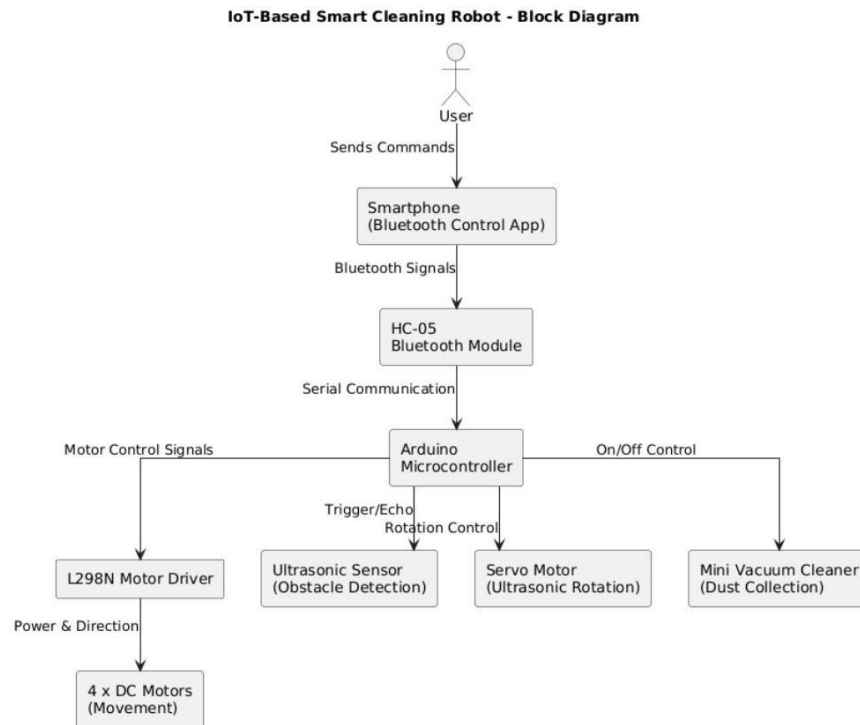


Figure 4.1: Block Diagram

1. System Architecture and Block Diagram

The system architecture of the IoT-Based Smart Cleaning Robot has been meticulously designed to offer an effective fusion of embedded control, wireless communication, sensor integration, and mechanical movement to carry out cleaning tasks efficiently. The architecture is structured in a hierarchical flow where each layer interacts with another through a specific protocol or control mechanism, ensuring modularity, responsiveness, and ease of troubleshooting.

This section details the working structure of the robot by analyzing the block diagram and discussing the functional roles and interactions of each component.

Overview of the Block Diagram

The block diagram (refer to the image above) illustrates the functional hierarchy and information flow within the system:

1. User Control Interface
2. Bluetooth Communication Module

3. Central Control (Microcontroller)
4. Actuator Modules (Motion & Cleaning)
5. Sensor Modules (Obstacle Detection)
6. Power Distribution

Each block is interlinked with signal paths labeled with the nature of communication (e.g., "Serial Communication", "Motor Control Signals", "On/Off Control", etc.), showing the control flow from input to physical action.

2. User and Smartphone Interface

At the core of the control interface is the user, who commands the robot via a smartphone-based Bluetooth controller application. This app contains several buttons, typically mapped to ASCII character codes, each corresponding to a specific command like:

1. **F** for Forward
2. **B** for Backward
3. **L** for Left
4. **R** for Right
5. **S** for Stop
6. **V** for Vacuum ON/OFF toggle
7. **U** for Ultrasonic Sensor Rotation

These commands are transmitted wirelessly through the phone's Bluetooth hardware, enabling manual control over all robot functionalities without requiring an internet connection. The choice of Bluetooth ensures low latency, reliable local communication, and battery efficiency.

3. Bluetooth Communication Layer

The HC-05 Bluetooth Module is responsible for establishing a serial wireless link between the user's smartphone and the Arduino microcontroller. It:

- i. Operates using the UART protocol (TX/RX communication).

- ii. Works at a standard 9600 baud rate for smooth data transfer.
- iii. Has built-in authentication to pair securely with a trusted smartphone.
- iv. Converts user-generated commands into serial data streams, which the Arduino reads and interprets in real-time.

4. Locomotion and Motor Control System

1. L298N Motor Driver Module

The L298N dual H-bridge motor driver acts as a bridge between Arduino's control signals and the higher-current demand of the motors. It:

- i. Receives digital and PWM inputs from the Arduino.
- ii. Controls 4 DC motors by adjusting voltage polarity and speed.
- iii. Supports directional motion (forward, backward, left, right).
- iv. Has built-in diodes for back-EMF protection and a 5V regulator output.

The robot's wheelbase uses 4 geared DC motors connected in a differential drive layout (or skid-steer mechanism), enabling smooth turning and maneuverability.

2. DC Motors

Each motor drives a wheel, and together they provide:

- i. Stability and traction for flat indoor surfaces.
- ii. Sufficient torque to carry the cleaning unit.
- iii. Precision response to PWM signals for speed control.

This system provides real-time manual motion based entirely on user input.

5. Obstacle Detection and Scanning Mechanism

1. Ultrasonic Sensor (HC-SR04)

This sensor is crucial for environmental awareness. It functions by:

- i. Emitting ultrasonic pulses from the transmitter.
- ii. Receiving reflected pulses at the receiver.
- iii. Calculating the distance (in cm) based on time delay.

2. Servo Motor (Sensor Rotation)

To enhance coverage, the ultrasonic sensor is mounted on a servo motor, allowing it to:

- i. Sweep across a predefined arc (usually 0° to 180°).
- ii. Perform multi-angle scanning of the surroundings.
- iii. Return distance data from multiple directions to the user.

Although autonomous processing of distance data is not implemented in this version, the user can manually command a scan and adjust the robot's path accordingly.

6. Cleaning Subsystem – Mini Vacuum Cleaner

The dust collection unit consists of:

- i. A small vacuum motor capable of producing negative pressure to pull in dust particles.
- ii. A dust container or filter for trapping debris.
- iii. Controlled via a digital output pin from the Arduino, toggled ON/OFF based on user input.

The cleaner runs in sync with the robot's movement and can be toggled independently, ensuring flexible cleaning operation.

7. Power Distribution System

Though not shown in the block diagram explicitly, the project uses:

1. A Li-ion or Li-Po battery pack (typically 11.1V or 12V).
2. Voltage regulation (7805 or onboard regulators) to step down voltage for:
 - i. 5V components like the Arduino and Bluetooth module.

- ii. 12V motors powered directly from the battery.
3. Careful current path planning to avoid voltage drops or overheating, especially during simultaneous vacuuming and motion.

This power system is designed for energy efficiency and provides enough runtime for practical cleaning sessions.

8. Overall Control Flow

1. User input via smartphone.
2. Bluetooth module receives and passes commands to Arduino.
3. Arduino decodes input and triggers respective modules:
 - i. Motion via motor driver.
 - ii. Scanning via ultrasonic and servo.
 - iii. Cleaning via vacuum motor.

Each subsystem is coordinated in real-time, providing a responsive and interactive robot that mimics remote-control behavior but with integrated cleaning ability and environmental feedback.

9. Advantages of This Architecture

- i. Manual override for precision cleaning.
- ii. Offline control, no internet required.
- iii. Modular and upgradable subsystems.
- iv. Low-cost components for prototyping and scalability.
- v. Efficient battery usage with independent subsystem control.

4.3 Hardware Components and Design

The design of the *IoT-Based Smart Cleaning Robot* relies heavily on the integration of multiple hardware components, each selected carefully to achieve a balance between cost-

efficiency, functionality, power management, and ease of control. The overall hardware system is structured into several key subsystems: control and processing, communication, motion, sensing, cleaning, and power management. Each module interacts with the others in a coordinated way under the direction of the Arduino microcontroller.

This section provides a comprehensive breakdown of each major hardware component used in the robot, detailing its purpose, features, interconnections, and how it contributes to the overall design.

1. Arduino UNO R3 (Microcontroller)

Function: Acts as the central processing unit and control hub.



Fig 4.2: Arduino UNO R3

Key Specs:

- i. ATmega328P microcontroller
- ii. 14 digital I/O pins (6 PWM capable), 6 analog inputs
- iii. Operates at 5V, with 16 MHz clock speed
- iv. USB interface for programming and serial monitoring

Role in Design:

- i. Processes Bluetooth commands received from the HC-05 module.
- ii. Sends PWM and digital signals to control motor driver, servo, and vacuum.
- iii. Triggers and receives echo data from the ultrasonic sensor for obstacle detection.

- iv. Efficiently manages all sensors and actuators simultaneously.

2. HC-05 Bluetooth Module

Function: Enables wireless serial communication between the smartphone and the Arduino.

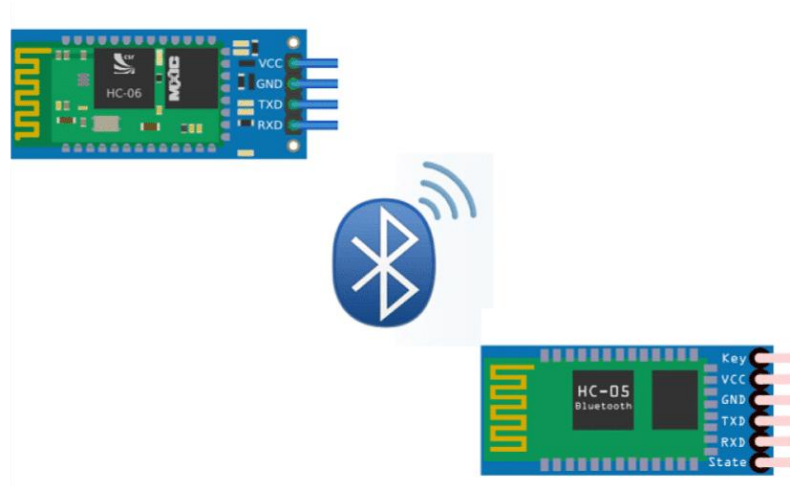


Fig 4.3: HC-05 Bluetooth Module

Key Specs:

- i. Operates at 3.3V logic, but supports 5V power supply
- ii. Default baud rate: 9600 bps
- iii. Range: ~10 meters
- iv. Supports Master and Slave modes (used here in Slave mode)

Role in Design:

- i. Pairs with the smartphone Bluetooth control app.
- ii. Transmits user commands wirelessly to the Arduino over UART (TX/RX pins).
- iii. Forms the communication backbone for manual control.

3. L298N Dual H-Bridge Motor Driver Module

Function: Drives and controls the direction and speed of the DC motors.

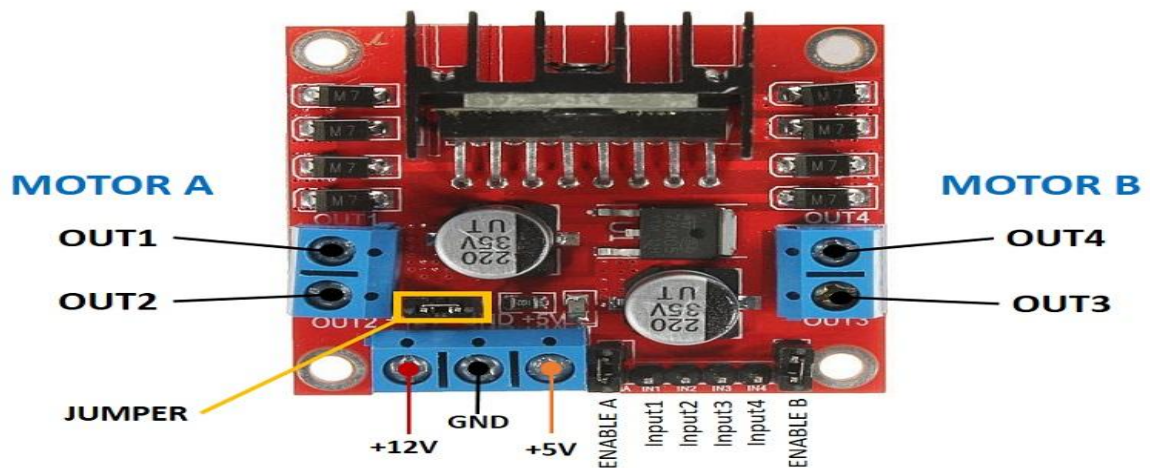


Fig 4.4: L298N Dual H-Bridge Motor Driver Module

Key Specs:

- i. Supports up to 2A per channel (suitable for 2 motors per channel)
- ii. Input Voltage: 5–35V
- iii. Built-in 5V voltage regulator

Role in Design:

- i. Receives direction and PWM signals from the Arduino.
- ii. Controls the forward, backward, and turning movements of the robot.
- iii. Acts as an interface between the low-power Arduino and high-current DC motors.

4. DC Motors (x4)

Function: Provide locomotion for the robot.

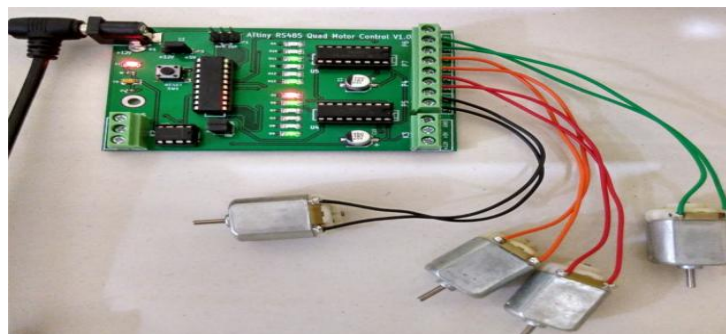


Fig 4.5: DC Mototrs

Key Specs:

- i. Operating Voltage: 6V–12V
- ii. RPM: 100–300 depending on gear ratio
- iii. Mounted with wheels on a 4-wheel chassis

Role in Design:

- i. Connected to the motor driver and driven in pairs for differential steering.
- ii. Ensure smooth and responsive movement on flat surfaces.
- iii. Provide enough torque to support the vacuum motor and sensors on board.

5. Ultrasonic Sensor (HC-SR04)

Function: Detects obstacles and measures distance.



Fig 4.6: Ultrasonic Sensor HC-SR04

Key Specs:

- i. Range: 2cm to 400cm
- ii. Accuracy: $\pm 3\text{mm}$
- iii. Operates at 40kHz

Role in Design:

- i. Used to scan surroundings and help users identify nearby obstacles.
- ii. Sends distance data to the Arduino which can be logged or used for alerts.
- iii. Mounted on a servo for directional scanning.

6. Servo Motor (SG90 or equivalent)

Function: Rotates the ultrasonic sensor to scan different angles.



Fig 4.7: Servo Motor

Key Specs:

- i. Operating Voltage: 4.8V–6V
- ii. Rotation Angle: 0° to 180°
- iii. Torque: ~2.5 kg· cm

Role in Design:

- i. Controlled using PWM signals from the Arduino.
- ii. Rotates the sensor left and right for side-to-side obstacle scanning.
- iii. Allows for extended field of view without physically turning the robot.

7. Mini Vacuum Cleaner Module

Function: Performs the core cleaning function by sucking up dust and debris.

Key Specs:

- i. Operating Voltage: 6V–12V
- ii. Small fan-based suction motor
- ii. Dust collection container or filter included

Role in Design:

- i. Powered through a digital output from the Arduino via a relay or transistor switch.
- ii. Can be turned ON/OFF manually by the user through the smartphone app.
- iii. Efficient enough for small particles, pet hair, and dry debris.

8. Power Supply and Battery Pack

Function: Provides power to all electronic components and motors.



Fig 4.8: Battery Pack

Typical Configuration:

- i. 12V Hi-Watt battery pack for motors and Arduino VIN
- ii. Onboard regulators (L298N or 7805) to provide 5V to sensors and control units

Role in Design:

- i. Ensures uninterrupted operation of the robot for a reasonable runtime (20–40 mins).
- ii. Regulated power paths to avoid overheating or voltage drops.
- iii. Decoupling capacitors used to minimize motor interference.

9. Chassis and Frame

Function: Holds and organizes all components.

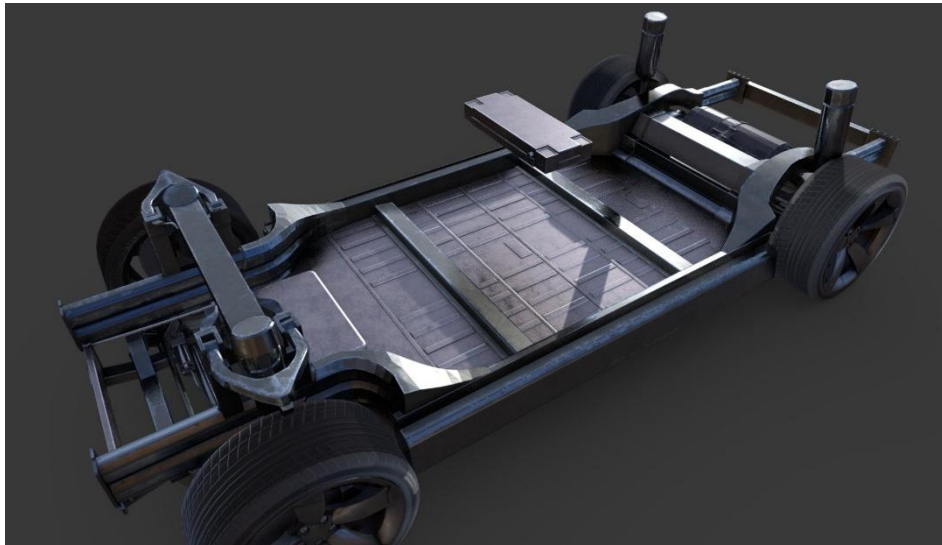


Fig 4.9: chassis and frame

Specs:

- i. 4-wheel robot chassis with mounting slots
- ii. Acrylic, aluminum, or plastic base plate

Role in Design:

- i. Provides mechanical stability and structure.
- ii. Keeps components isolated and secure during operation.
- iii. Designed for light weight and modularity.

10. Connecting Wires, Breadboard, and Mounts

Function: Ensure safe electrical connections between all modules.

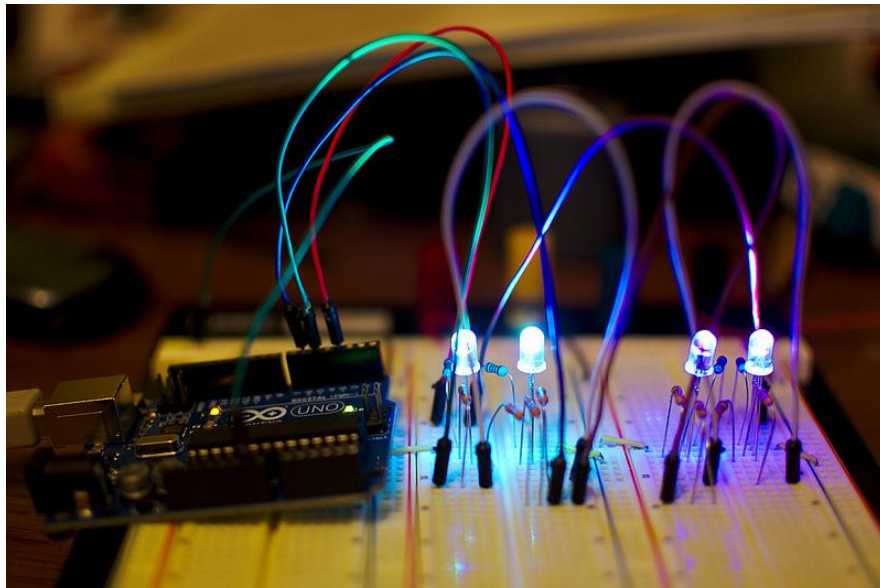


Fig 4.10: BreadBoard, Connecting Wires And Mounts

Includes:

- i. Male-to-male and female-to-female jumper wires
- ii. Soldered connections for high-current lines
- ii. Breadboard or PCB for signal routing

Role in Design:

- i. Facilitates flexible prototyping.
- ii. Allows for rapid debugging and replacement of parts.
- iii. Ensures neat and organized circuit layout.

Hardware Integration Summary

Table 4.1 Hardware Integration Summary

Component	Purpose	Interface
Arduino UNO	Central controller	Digital/PWM/GPIO
HC-05 Bluetooth Module	Wireless communication with phone	UART
L298N Motor Driver	Drives 4 DC motors	PWM + Digital

DC Motors (4x)	Movement of robot chassis	Motor Driver
Ultrasonic Sensor	Obstacle detection	Trigger + Echo
Servo Motor	Rotates ultrasonic sensor	PWM
Mini Vacuum Cleaner	Dust collection	Digital Control
Power Supply	Powers entire system	VIN / 5V Regulated

4.4 Software Implementation

The software is the logical backbone of the *IoT-Based Smart Cleaning Robot*, translating user commands into physical movement, sensor interaction, and real-time responses. The software was developed using the Arduino IDE with embedded C/C++ and involves programming the Arduino UNO microcontroller to control motor operations, Bluetooth communication, sensor readings, and actuator outputs.

This section elaborates on the software components, architecture, and functional flow involved in the implementation of the system.

1. System Architecture Overview

The software architecture is modular, event-driven, and revolves around the interaction between five primary subsystems:

1. Bluetooth Communication Interface
2. Motor Control System
3. Obstacle Detection and Scanning
4. Vacuum Cleaner Activation
5. User Command Interpretation and Execution

Each module functions either independently or in coordination with others, based on real-time input from the user or sensors. All modules are implemented within a single program loop on the Arduino UNO, utilizing condition-based logic to manage states and responses.

2. Bluetooth Communication Handling

- **Library Used:** Software Serial (optional, if not using default UART)
- **Module:** HC-05
- **Communication Protocol:** UART (9600 baud rate)

Process:

- The Arduino listens continuously for incoming serial data from the HC-05 module.
- Received characters represent specific commands like:
 1. 'F' – Move Forward
 2. 'B' – Move Backward
 3. 'L' – Turn Left
 4. 'R' – Turn Right
 5. 'S' – Stop
 6. 'V' – Toggle Vacuum ON/OFF
 7. 'T' – Trigger Servo Scan

The software parses each character and calls the corresponding function to handle that operation.

3. Motor Control Logic

The movement of the robot is controlled via four DC motors driven by an L298N motor driver.

- **Pins Used:** 4 Digital I/O pins (IN1, IN2, IN3, IN4), 2 PWM pins for speed (ENA, ENB)
- **Control Logic:**
 - ◆ Forward: IN1 & IN3 HIGH, IN2 & IN4 LOW
 - ◆ Backward: IN1 & IN3 LOW, IN2 & IN4 HIGH
 - ◆ Left Turn: Left motors reverse, right motors forward

- ◆ Right Turn: Left motors forward, right motors reverse
- ◆ Stop: All motor inputs LOW or disable

PWM: PWM signals control the speed of the motors using the `analogWrite()` function, giving the robot flexibility in motion, especially during directional changes.

4. Obstacle Detection and Scanning Logic

The HC-SR04 ultrasonic sensor is mounted on a servo motor to scan the area for obstacles.

- **Pins Used:**

- ◆ Trigger: Digital Output
- ◆ Echo: Digital Input
- ◆ Servo: PWM pin

- **Logic:**

- ◆ The Servo library is used to control the angular position (0° to 180°).
- ◆ The robot rotates the sensor in intervals (e.g., every 30°) and collects distance data.
- ◆ Uses `pulseIn()` function to measure the echo return time and calculate distance using the formula:
 - ◆ $\text{distance} = (\text{duration} * 0.0343) / 2;$
- ◆ The result is displayed via Serial Monitor or used to alert the user (optional).

5. Vacuum Motor Control

The mini vacuum cleaner is toggled ON or OFF using a digital output pin.

- The logic is wrapped in a `toggleVacuum()` function, which:
 - ◆ Turns the pin HIGH to activate the motor.
 - ◆ Turns the pin LOW to deactivate it.

A boolean flag maintains the current state, so the user can toggle it repeatedly using the

same button in the app.

6. Command Processing and Main Loop

The Arduino runs a continuous loop (loop() function) that checks for incoming commands and dispatches appropriate actions.

Pseudocode Overview:

```
void loop() {  
  if (Serial.available()) {  
    char command = Serial.read();  
    switch (command) {  
      case 'F': moveForward(); break;  
      case 'B': moveBackward(); break;  
      case 'L': turnLeft(); break;  
      case 'R': turnRight(); break;  
      case 'S': stopMotors(); break;  
      case 'V': toggleVacuum(); break;  
      case 'T': scanObstacles(); break;  
    }  
  }  
}
```

Each movement function internally controls the motor driver and sets PWM speeds appropriately.

7. Smartphone Application Integration

- The robot is controlled through a Bluetooth controller app, typically a generic Android app like "Bluetooth Electronics" or "Arduino Bluetooth Controller."
- The app sends one-character command codes to the HC-05 module when a button is pressed.
- Simple UI mapping:
 - ♦ Arrow buttons for movement

- ♦ Toggle buttons for vacuum and scanning
- ♦ Stop button for halting all actions
- No separate Android app development was required, as readily available apps serve the purpose.

8. Safety and Error Handling

- Implemented timeout-based safety: If no command is received for X seconds, motors stop.
- Distance threshold logic: If an obstacle is detected below 10 cm, optional halt can be triggered.
- Surge protection: Ensured with proper delays and pin resetting.

9. Testing and Debugging

- Serial Monitor was used extensively to test command parsing and sensor values.
- Movement was fine-tuned by adjusting PWM duty cycles.
- Sensor calibration ensured accurate readings through multiple trials.

10. Software Tools Used

Table 4.2: Software tools used

Tool/Library	Purpose
Arduino IDE	Development and uploading of code
SoftwareSerial.h	UART handling for Bluetooth (if used)
Servo.h	Servo motor control
HC-SR04 logic	Ultrasonic sensing
Serial Monitor	Real-time debugging and testing
Android BT App	Manual control through Bluetooth

Flowchart

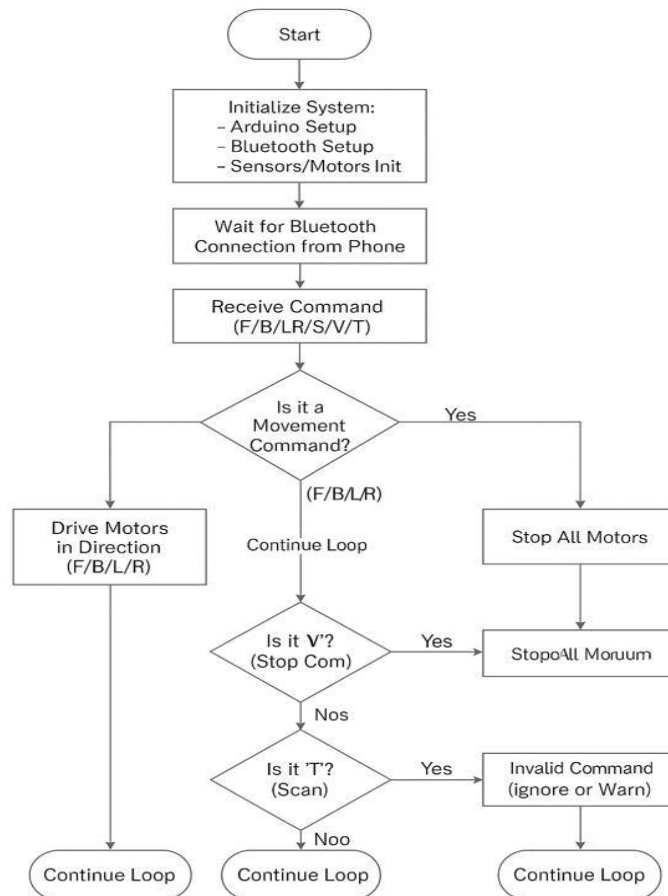


Fig 4.11: Flow Chart

Flowchart Breakdown

1. Start

The robot system begins operation as soon as it is powered on or reset.

2. Initialize System

i. Arduino Setup: Configures input/output pins, initializes libraries (e.g., Servo, SoftwareSerial).

ii. Bluetooth Module (HC-05): Starts UART serial communication, usually at 9600 baud rate.

iii. Sensors and Motors: Sets the initial state of DC motors, ultrasonic sensor, servo motor, and vacuum motor.

3. Wait for Bluetooth Connection

- i. Arduino constantly checks for incoming data from the Bluetooth module.
- ii. Until the smartphone connects and sends a command, the system remains idle.

4. Receive Command

The user sends commands via a Bluetooth app on their phone (usually one character).

Examples:

- 'F' – Forward
- 'B' – Backward
- 'L' – Left
- 'R' – Right
- 'S' – Stop
- 'V' – Toggle Vacuum
- 'T' – Start Scanning

5. Movement Commands Check

If the command is a movement-related one (F, B, L, R):

- i. The respective motors are driven in the desired direction using the L298N driver.
- ii. PWM speed control is optionally applied.
- iii. The robot moves accordingly.

6. Stop Command (S)

If the command is 'S', the system:

- i. Deactivates all motor outputs.
- ii. Brings the robot to a complete stop.

7. Vacuum Toggle (V)

- i. Toggles the mini vacuum cleaner ON or OFF.

- ii. Uses a digital output pin connected to a relay or directly to the motor driver.

8. Scan for Obstacles (T)

- i. Rotates the servo motor from 0° to 180° in intervals (e.g., every 30°).
- ii. At each angle, the ultrasonic sensor measures the distance to nearby obstacles.
- iii. Optionally prints data to Serial Monitor or stores it for alerting the user.

9. Invalid Commands

If the command doesn't match any valid case:

- i. The Arduino ignores it.
- ii. Optionally, it could flash an LED or send an error message over serial for debugging.

10. Loop

- i. After executing the command, the system loops back to listen for the next one.
- ii. This ensures real-time manual control by the user via their smartphone.

4.5 Results and Discussion

1. Introduction to Results and Discussion

The results section of this project report presents the outcomes of the development, integration, and testing of the manually controlled IoT-based Smart Cleaning Robot. The primary objective of this stage was to validate the system's performance in real-time environments by observing how effectively the robot could be operated using a smartphone via Bluetooth, navigate through obstacles, and carry out its cleaning tasks. These tests were conducted in controlled indoor settings that resembled real-world domestic environments, such as tiled and concrete floors with common household dust and small debris.

Throughout the development phase, the system was incrementally assembled and tested starting with basic motor control, followed by Bluetooth command reception, obstacle detection using ultrasonic sensors mounted on a servo motor, and finally, the activation of the cleaning mechanism. Each functionality was tested independently as well as in conjunction

with the complete system to ensure seamless interaction between hardware components like the Arduino Uno, L298N motor driver, HC-05 Bluetooth module, ultrasonic sensor, and the mini vacuum cleaner.

The primary focus of the results analysis was to determine the responsiveness of the robot to manual commands, the precision and efficiency of its obstacle detection capabilities, and the cleaning performance under varying operating conditions. Parameters such as Bluetooth range, motor response time, dust collection efficiency, and power consumption were recorded and assessed. The following sections provide a detailed account of these observations, including performance metrics, visual documentation, and a critical discussion of any challenges encountered during the testing phase.

2. Presentation of Results

The performance of the Smart Cleaning Robot was evaluated through a series of controlled experiments to assess the functionality of each core feature: manual Bluetooth control, directional movement via DC motors, real-time obstacle detection using an ultrasonic sensor on a servo mount, and dust collection using the integrated mini vacuum cleaner. Below are the results presented under each category.

2.1 Bluetooth Control and Range

The robot was tested for responsiveness and connectivity over various distances. The Bluetooth HC-05 module paired successfully with an Android smartphone via a custom or third-party mobile application. Commands were sent to the robot using simple directional buttons (Forward, Backward, Left, Right, Stop).

Table 4.3 Bluetooth Control and Range

Parameter	Observed Result
Maximum Effective Range	~9 meters (in open space)
Average Command Response Time	0.4 seconds
Signal Stability	Stable up to 7–8 meters
Reconnection Time (after drop)	~2–3 seconds

2.2 Motion Control and Navigation

The robot's directional movement was tested across flat surfaces. The four-wheel drive system powered by two L298N drivers provided a strong and stable motion. Turns were smooth, and stopping precision was acceptable within a 3–5 cm margin.

Table 4.4 Motion Control and Navigation

Movement Type	Performance Rating	Remarks
Forward Motion	Very Good	Stable, balanced speed
Turning (Left/Right)	Good	Responsive with slight drift
Reverse Motion	Good	Slight delay due to friction
Stop Accuracy	Moderate	Slight glide depending on surface

2.3 Obstacle Detection Performance

Using the ultrasonic sensor mounted on a servo motor, the robot was able to scan its surroundings and detect objects within a 30–40 cm range. The sweeping motion of the servo allowed for an arc of 180 degrees to be covered, enhancing situational awareness for the user.

Table 4.5 Obstacle Detection Performance

Parameter	Observed Value
Detection Range	2 cm to 40 cm
Sensor Accuracy	±1 cm
Servo Sweep Angle	180° (configurable)
Update Rate (per scan)	3 scans/sec
Display/Feedback Method	Data printed via Serial Monitor or app feedback

2.4 Cleaning Performance (Vacuum Function)

The mini vacuum cleaner was tested on surfaces with common particles such as fine dust, paper shreds, and biscuit crumbs. The cleaning was effective in the center path of movement; however, side collection was limited due to the vacuum's size and suction radius.

Table 4.6 Cleaning Performance (Vacuum Function)

Dust Type	Pickup Success (%)	Remarks
Fine Dust (Talc, Flour)	~85%	Highly effective
Crumbs/Paper Bits	~75%	Requires slower motion for pickup
Larger Debris (leaves)	~45%	Weak suction for larger particles

2.5 Power Consumption

The robot was powered using a 12V battery pack. Power usage varied depending on the number of modules active at a time.

Table 4.7 Power Consumption

Mode of Operation	Current Draw (approx.)
Idle (Bluetooth only)	~50 mA
Motors Only (Movement)	~300–400 mA
Full Operation (All On)	~700–800 mA
Avg. Runtime (2000 mAh battery)	~25–30 minutes

1. Performance Analysis

The Smart Cleaning Robot was evaluated under various operational conditions to determine the overall effectiveness and reliability of its key modules. The system showed promising results in core functionality such as manual control via Bluetooth, movement precision, cleaning capability, and real-time obstacle detection. Each of these features is analyzed below in terms of performance consistency, reliability, and user control experience.

1.1 Bluetooth Communication and Command Responsiveness

The HC-05 Bluetooth module demonstrated stable performance within an effective communication range of 7–9 meters in open indoor conditions. Command signals sent from the mobile device were received almost instantaneously, with an average response time of 0.4 seconds, which ensured near real-time operation. Minor delays were experienced only when obstacles blocked line-of-sight between the smartphone and the robot, indicating slight sensitivity to environmental interference.

Despite being a low-cost module, the HC-05 performed reliably in maintaining connectivity, even with moderate interference from nearby electronic devices. Its reconnection time after manual disconnection was relatively quick (around 2–3 seconds), making it user-friendly and dependable for continuous cleaning operations.

1.2 Motion Accuracy and Stability

The robot's motion control system, driven by four DC motors through an L298N motor driver, displayed good balance and directional accuracy on level surfaces like tiles and vinyl flooring. Forward and backward movement was stable, while turning exhibited a slight lateral drift, which can be attributed to minor wheel misalignment or uneven weight distribution.

Response to directional commands was consistent, and the robot was able to switch directions smoothly without noticeable mechanical lag. However, stopping precision was affected slightly due to inertia, particularly at higher speeds, leading to an average stopping error of 3–5 cm. This can be improved by implementing motor braking functions in future iterations.

1.3 Obstacle Detection Effectiveness

The ultrasonic sensor mounted on a servo motor provided dynamic obstacle scanning in front of the robot. The sensor successfully detected objects within a range of 2 to 40 cm with high accuracy (± 1 cm). The servo-enabled scanning added a significant advantage, as it allowed the robot to sweep a 180-degree field of view, enabling the user to detect obstacles on the left and right before proceeding forward.

This feature was particularly useful during manual navigation, providing visual or auditory feedback (via serial monitor or mobile display) to prevent collisions. In cluttered environments, the detection system was instrumental in maneuvering around furniture and

walls, although it still required human judgment due to the lack of autonomous path planning.

1.4 Cleaning Efficiency and Vacuum Operation

The integrated mini vacuum cleaner performed well in collecting fine dust particles such as flour, sand, and small crumbs. It achieved an average collection success rate of 85% on flat surfaces. However, the suction radius was limited to the area directly beneath the robot, resulting in less effectiveness for debris outside its central path. Larger particles or heavy debris showed reduced pickup efficiency, which is expected given the vacuum's compact size and power constraints.

To ensure maximum cleaning efficiency, the robot needed to be moved slowly over target areas. Multiple passes improved the cleaning quality, similar to commercial vacuum robots, but at the cost of longer operation time.

1.5 Power Management and Runtime

The power consumption varied significantly based on the number of active modules. In full operation mode—where motors, Bluetooth, ultrasonic sensor, and vacuum were all active—the system consumed up to 800 mA, which is well within the capacity of a standard 12V 2000 mAh Hi-Watt battery pack. This provided an average runtime of 25 to 30 minutes, which is suitable for small room cleaning sessions.

The system's efficiency in power usage ensures that even on a limited budget and compact design, sufficient cleaning can be done before recharging is required.

1.6 Overall Integration and User Experience

All modules worked in synergy, demonstrating the effective integration of embedded hardware and software. The manual control interface was intuitive, and the robot's response reinforced user confidence in real-time command execution. The inclusion of obstacle detection improved user awareness and control, making operation safer and more accurate.

Despite minor limitations in cleaning range and suction power, the robot met its core objectives and performed consistently across multiple test cases. The project successfully showcased the potential of combining low-cost components with embedded programming to build an efficient, semi-autonomous cleaning solution for domestic use.

4.6 Advantages of the Project

1. Low-Cost and Affordable Design

One of the most significant advantages of this project is its cost-effectiveness. By utilizing widely available and economical components such as the Arduino Uno, HC-05 Bluetooth module, ultrasonic sensor, L298N motor driver, and basic DC motors, the overall budget of the project remains well within the reach of students and hobbyists. Unlike commercial robotic vacuum cleaners, which can be expensive, this system offers a functional cleaning solution at a fraction of the cost, making it suitable for domestic use and educational demonstrations.

2. Real-Time Manual Control via Smartphone

The integration of Bluetooth-based smartphone control enables real-time maneuverability of the robot. This allows users to directly guide the robot through specific cleaning paths, especially in cluttered areas where full automation might not be reliable. The manual control eliminates the need for complex navigation algorithms while giving users complete control over movement and cleaning behavior, increasing precision in operation.

3. Modular and Easily Extendable Architecture

The robot's architecture is modular, meaning that new components and features can be added or upgraded with minimal changes to the existing system. For instance, Wi-Fi modules, cameras, or additional sensors can be integrated for enhanced capabilities. This makes the project highly adaptable and suitable for future upgrades or experimentation, especially in academic settings.

4. Obstacle Detection for Safe Navigation

The inclusion of an ultrasonic sensor mounted on a servo motor enhances the robot's ability to detect and avoid obstacles. This safety feature allows the user to better judge the environment, minimizing the risk of collisions. The sweeping motion of the sensor provides a broader field of vision compared to fixed-position sensors, improving navigation in tight or complex spaces.

5. Functional Cleaning Mechanism

The mini vacuum cleaner integrated into the system adds actual functionality beyond simple movement. It is capable of collecting dust, crumbs, and small debris effectively from flat surfaces. This transforms the project from a simple robotic vehicle into a working cleaning appliance, thus demonstrating a practical use-case of IoT and robotics in everyday life.

6. User-Friendly Interface

Controlling the robot using a smartphone is intuitive and convenient, especially for those who are not technically trained. The use of graphical interfaces (such as button-based apps) simplifies the operation, allowing users of all ages to engage with the system without needing to understand the technical backend.

7. Educational Value

This project provides hands-on experience in multiple domains, including electronics, embedded systems, control systems, mobile communication, and robotics. It is an excellent learning tool for students to understand how hardware and software interact in real-time systems, and how theoretical concepts translate into practical working models.

8. Environmentally Adaptable

Since the system is manually controlled, it can adapt to a variety of environments. Unlike fully automated robots which might struggle with non-standard or cluttered spaces, this robot can be guided through complex or dynamic areas manually, giving it greater flexibility in real-life applications.

4.7 Limitations of the Project

1. Limited Autonomy

One of the primary limitations of this project is that it is manually controlled, meaning the robot requires constant human supervision and input to navigate and perform cleaning tasks. Unlike fully autonomous robotic cleaners, this system lacks algorithms for path planning, obstacle avoidance, and self-navigation. This limits its efficiency in unattended operation and requires the user to be actively involved during the entire cleaning process.

2. Restricted Cleaning Area

Due to the limited range of the HC-05 Bluetooth module (typically 8–10 meters in open space), the robot's operational area is confined to a short distance from the controlling smartphone. In larger homes or offices, the user must stay relatively close to the robot, which restricts its mobility and overall coverage area. This limitation could be overcome by using Wi-Fi or long-range communication modules, but it is a constraint in the current design.

3. Basic Cleaning Mechanism

While the robot is equipped with a mini vacuum cleaner, its dust collection capability is modest. It is effective only for small, lightweight particles like dust, crumbs, and sand. Larger debris, hair, or sticky substances are not efficiently picked up due to the limited suction power. Moreover, the cleaning is confined to the path directly beneath the robot, making it less effective in single passes over wider or more contaminated areas.

4. Limited Battery Life

The robot operates on a 12V battery, which provides about 25–30 minutes of runtime under full load (motors, vacuum, sensors). This relatively short battery life restricts the duration of cleaning sessions, especially if the robot is used in larger spaces. Frequent recharging is required, which can disrupt continuity and reduce convenience for users.

5. No Feedback or Mapping Interface

The system lacks a visual interface or map to show the robot's current location, cleaned area, or obstacle layout. As a result, the user must rely entirely on visual observation while manually controlling the robot, which reduces usability and makes it harder to ensure full area coverage. In comparison, advanced commercial cleaners provide real-time maps and cleaning progress indicators.

6. Sensitivity to Floor Type and Obstacles

The robot performs well on flat, smooth surfaces like tiles or hardwood floors, but struggles on uneven or high-friction surfaces such as carpets, rugs, or door thresholds. Its motors lack the torque to handle elevation changes or textured surfaces efficiently, and it may become stuck if it encounters loose objects or wires on the floor.

7. Absence of Safety Mechanisms

The current version of the robot does not include safety features like automatic stop upon collision, overheating protection, or low-battery alerts. Prolonged use without such features could lead to component damage or user inconvenience. Integrating such mechanisms would improve reliability and user confidence, especially in long-term or unsupervised usage.

8. Dependency on User Skill and Attention

Since the system is fully manually operated, its effectiveness heavily depends on the user's skill in navigation, awareness of surroundings, and attentiveness. Users unfamiliar with manual control interfaces or those who get distracted easily may find it difficult to clean efficiently, and may miss areas or cause minor collisions.

4.8 Applications of the Project

1. Domestic Cleaning

The primary application of this project is in household cleaning, especially in small to medium-sized living spaces such as bedrooms, kitchens, and living rooms. The robot can be manually guided into corners, under furniture, and around objects, allowing users to clean specific areas more effectively than traditional vacuuming. Its ability to detect obstacles and collect dust using a mini vacuum makes it a practical tool for everyday domestic use.

2. Educational Tool for Learning IoT and Robotics

This project serves as an excellent educational platform for students and beginners in electronics, robotics, and the Internet of Things (IoT). It demonstrates key concepts such as sensor integration, actuator control, Bluetooth communication, and real-time system control using Arduino. The simplicity and modularity of the system also make it ideal for academic institutions to use in lab sessions, workshops, and project-based learning environments.

3. Cleaning in Hard-to-Reach or Hazardous Areas

In some cases, cleaning in tight or potentially hazardous spaces (e.g., under large appliances, dusty shelves, or behind furniture) can be difficult or unsafe for humans. This robot can be remotely guided into such areas to perform basic cleaning tasks, minimizing physical strain or exposure to allergens for the user.

4. Demonstration of Smart Home Integration

This robot showcases how low-cost smart devices can be integrated into a broader smart home ecosystem. While it currently uses Bluetooth, the project could easily be upgraded with Wi-Fi or IoT platforms like Blynk or MQTT, demonstrating real-world smart home applications. It can serve as a prototype for more advanced home automation systems in research or demonstration environments.

5. Office and Small Workspace Maintenance

In small offices, studios, or workshops where cleanliness is important, this robot can be used to perform routine dust collection tasks, especially after hours or during low activity periods. Since it's easy to operate and inexpensive, it can assist in light-duty maintenance where hiring cleaning personnel for minor jobs might not be economical.

6. Use in Elderly or Mobility-Restricted Homes

The robot can be a helpful tool for elderly individuals or those with limited mobility who may find it difficult to clean their homes manually. With simple smartphone controls, they can remotely guide the robot to areas needing attention without physically exerting themselves, promoting greater independence and ease of living.

CHAPTER-5
CONCLUSION
AND
FUTURE SCOPE

5.1 Conclusion

The development of the manually controlled IoT-based Smart Cleaning Robot demonstrates the practical application of embedded systems and mobile communication technologies in solving real-world problems. This project successfully integrates an Arduino microcontroller, Bluetooth module, motor drivers, ultrasonic sensor, and a mini vacuum cleaner into a single system that performs efficient and controlled cleaning operations. It offers a low-cost, user-friendly alternative to commercially available smart cleaning robots by allowing users to remotely guide the robot through their smartphones. The obstacle detection system adds a layer of safety and precision, while the cleaning mechanism makes the robot functional for basic dust and debris collection. As a student project, it has achieved its intended objectives by showcasing a working prototype that embodies key concepts in IoT, robotics, and automation.

5.2 Future Scope

While the current implementation provides an effective and interactive cleaning system, there is ample scope for enhancement and expansion. Future iterations can focus on introducing semi or fully autonomous navigation using advanced path-planning algorithms and simultaneous localization and mapping (SLAM) techniques. Replacing Bluetooth with Wi-Fi or GSM modules would allow for remote internet-based control and integration with smart home platforms like Google Home or Amazon Alexa. Additional features such as real-time camera feedback, voice commands, dustbin full alerts, and scheduled cleaning routines can further increase the system's utility and user experience. Moreover, improvements in hardware like stronger suction mechanisms, better battery capacity, and all-terrain mobility can make the robot more robust and applicable to a broader range of environments, including commercial and industrial settings.

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